



RV College of
Engineering®

MLOPs Unit 5 flipped class

Governance in different countries - ROLEPLAY

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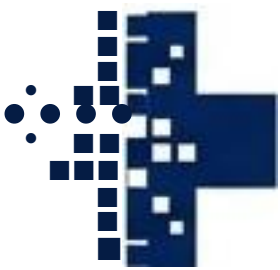
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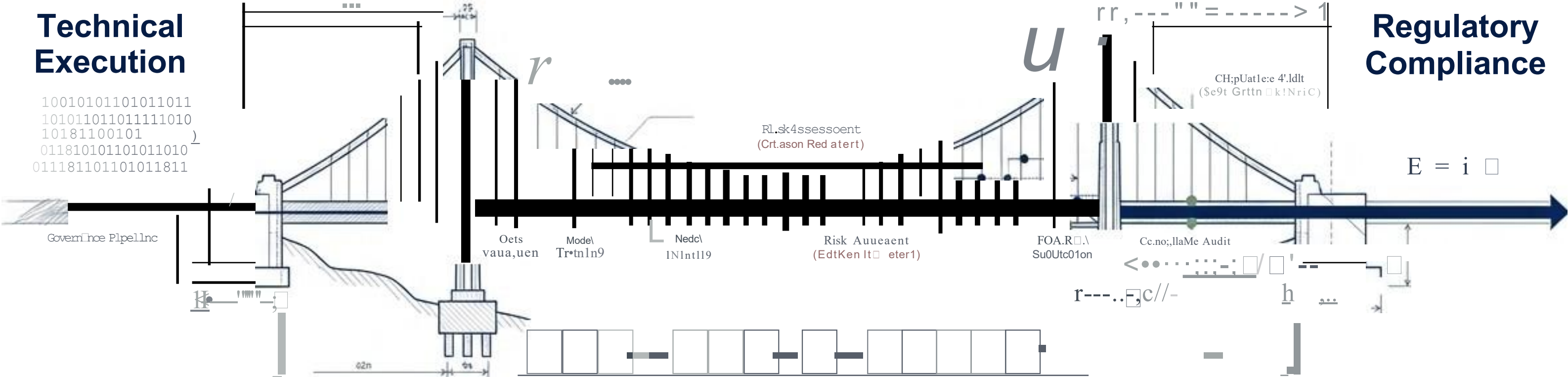
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Vital-Scan AI: A Strategic Governance Playbook for Global Launch

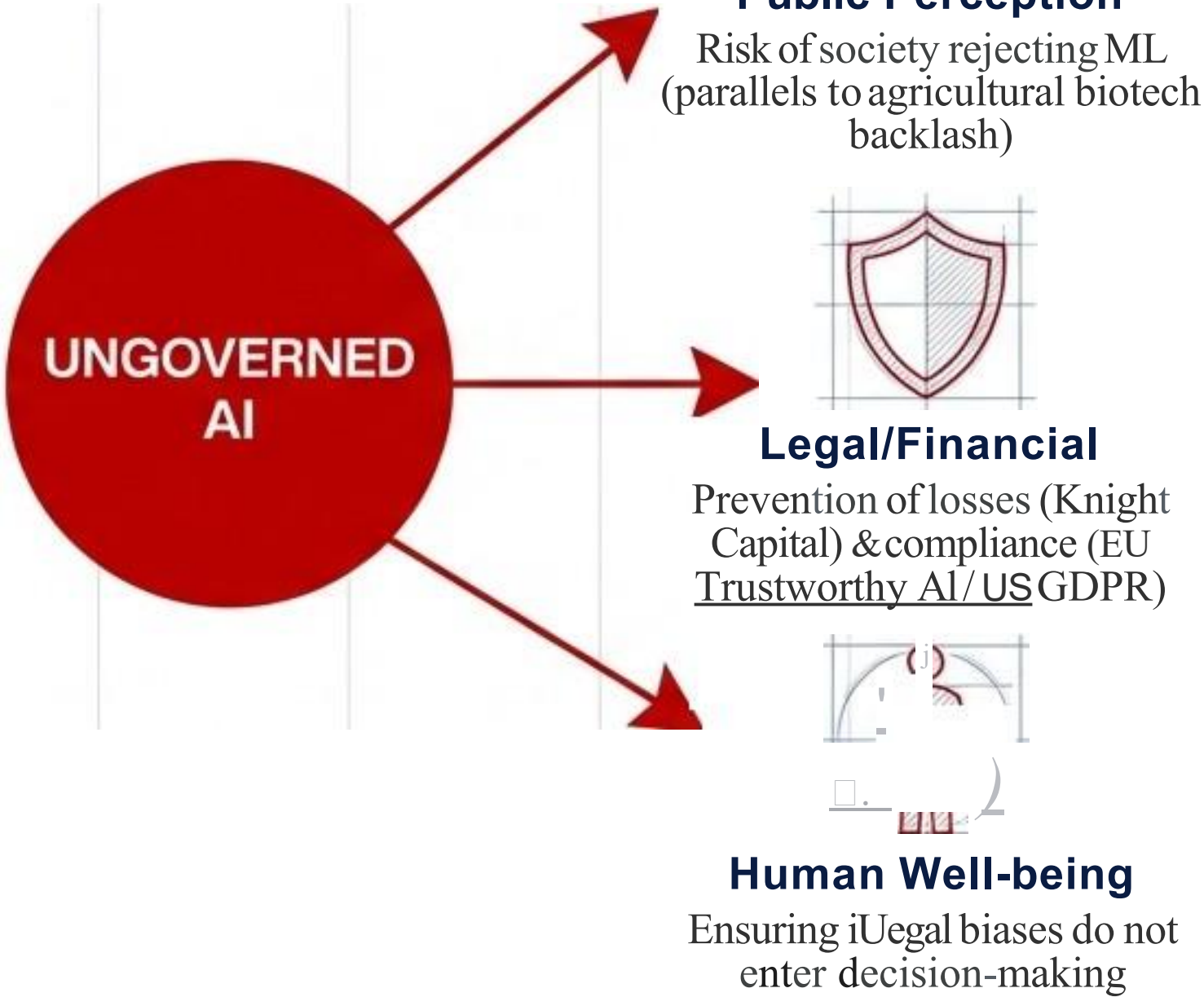
Operationalizing the 8-Step MLOps Governance Framework for High-Stakes Medical Diagnostics



We face an Opaque Technology Risk that threatens deployment.

Governance Pipeline

The core problem is "Opaque Technology Risk": The tendency of ML systems to invent decision processes that are hard to understand, leading to "accidental harm" or bias if left ungoverned.



THE SOLUTION: A unified, cross-functional 8-Step Governance Framework.

History proves that a lack of governance leads to catastrophic failure.

Governance Pipeline - - - - - □

The Trust Failure



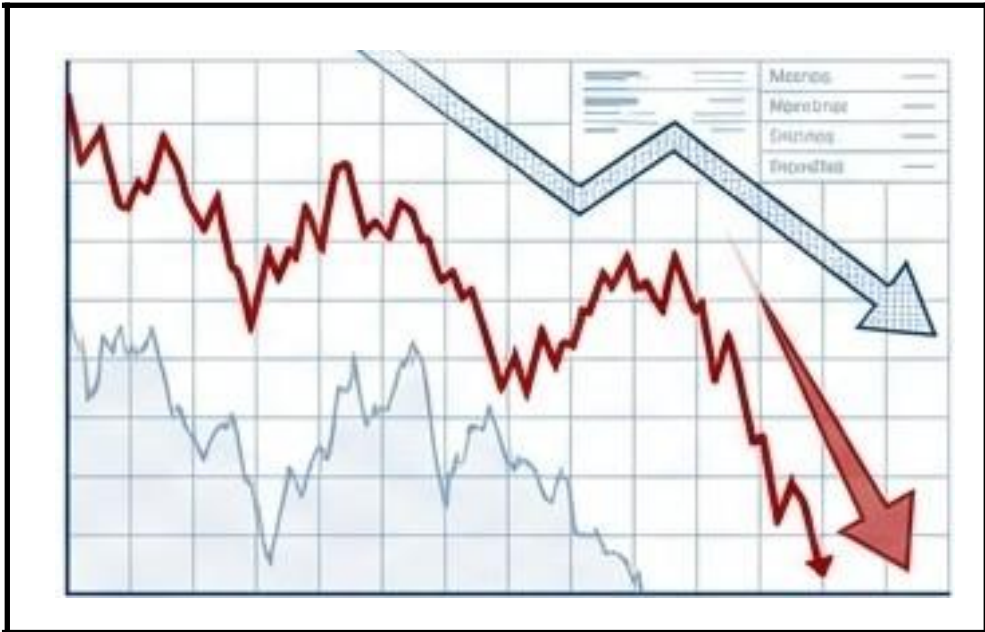
Merriweather Regular
Lesson: Without public trust, the product dies.

The Bias Failure



Amazon's Recruitment Tool
Lesson: Good intentions can still lead to automated discrimination.

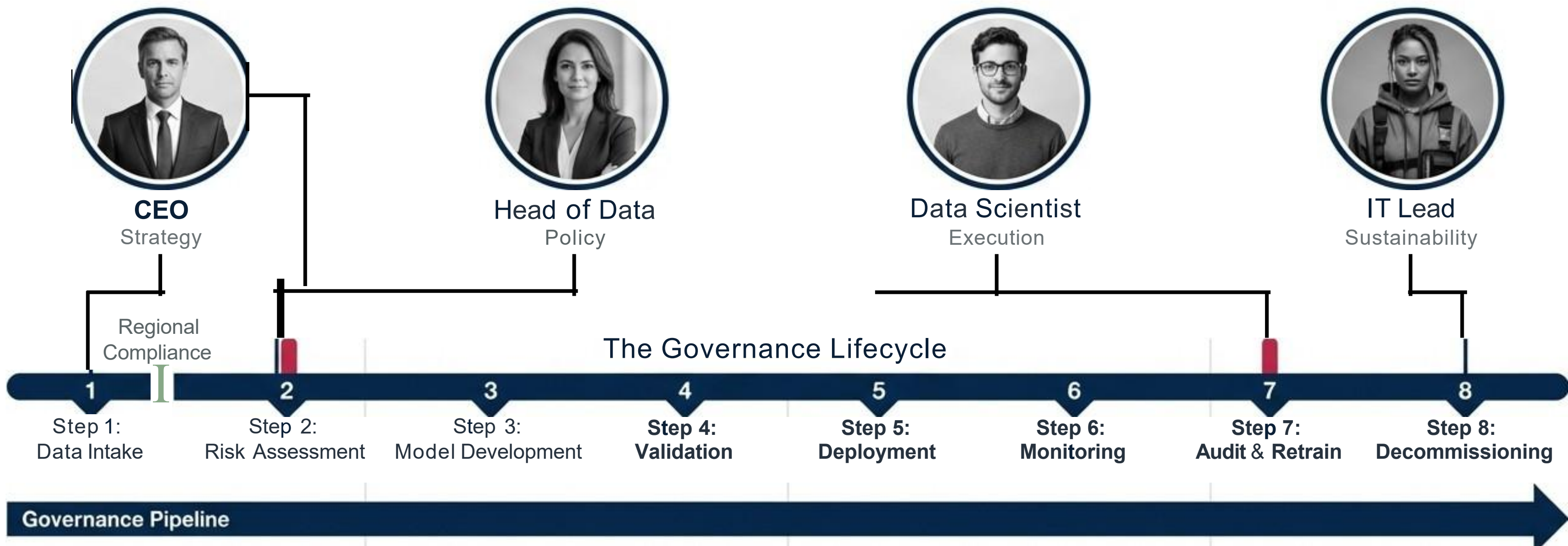
The Operational Failure



Knight Capital
Lesson: Ungoverned deployment can cause massive financial loss in minutes.

Key Takeaway: Vital-Scan AI must avoid these specific pitfalls through proactive role-based governance.

Four key stakeholders own the accountability for the 8-Step Framework



This is not just an IT project. It is a cross-functional mandate bridging the gap between the boardroom and the server room.

PERSONA CARD



Role: The CEO

Focus: Accountability & Strategy

Metric: Public Trust

The CEO sets the ethical compass to safeguard public trust.



"We've seen what happens when public opinion turns. Without public trust, our medical AI will never materialize."

- Core Responsibility (Step 2): Establishing the Ethical Position.
- Human-Centered Approach: A tool for doctors, not a replacement for judgment.
- Risk Avoidance: Active audit for bias to prevent reputation damage.

PERSONA CARD



Role: Head of Data (CDO)

Focus: Policy & Regulation

Metric: Compliance

The Head of Data navigates the divergence between US and Singapore markets.

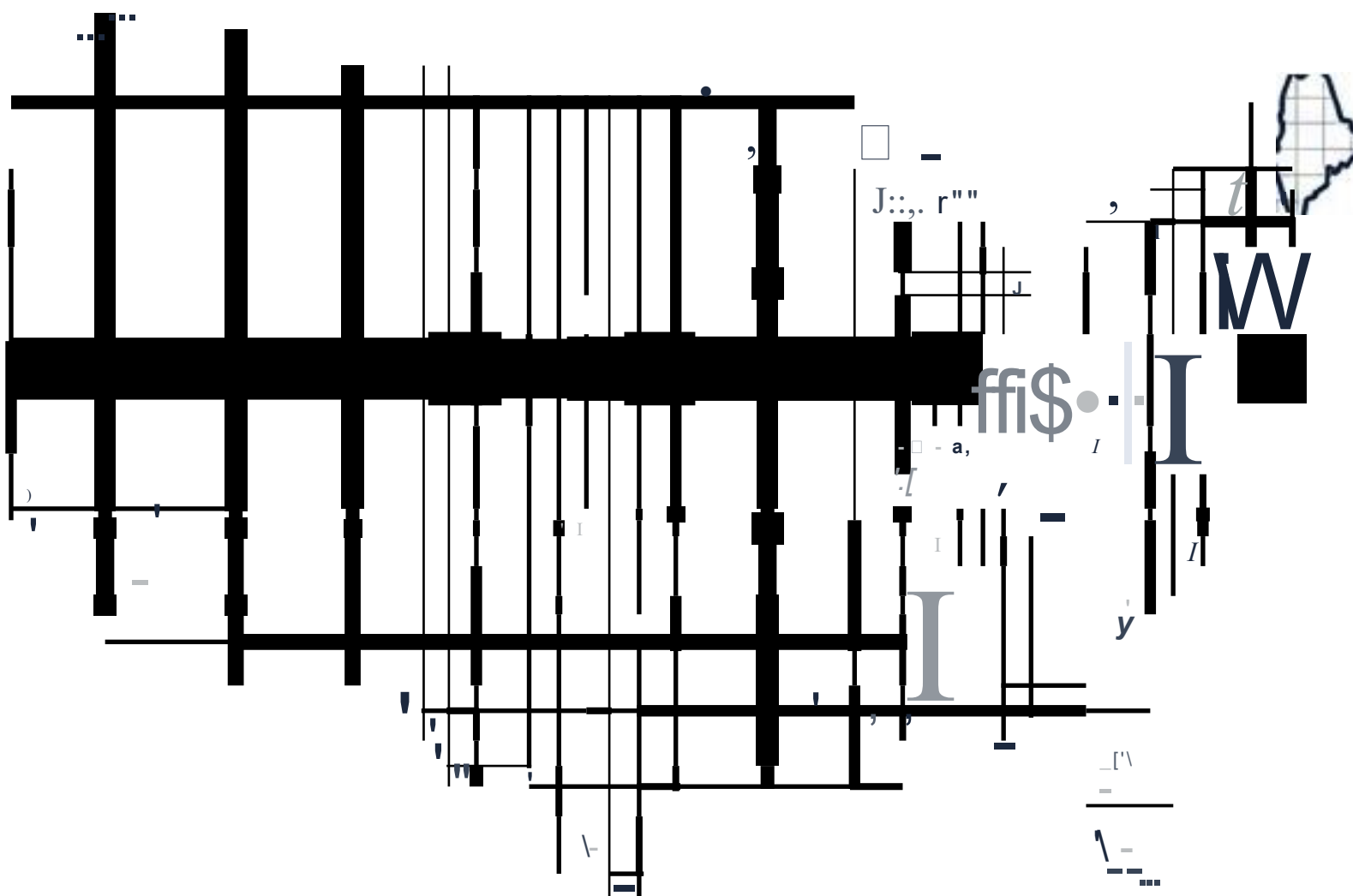


The Strategic Shift: Moving past paper forms to a Centralized Model Registry. This ensures every version is traceable if a diagnosis goes wrong.

The United States requires a strict, process-based approach to traceability.



Governance Pipeline



Regulatory Framework: Sectoral & Decentralized
Managed by specific agencies (FDA)

Key Guidelines: FDA's GxP & NIST AI Risk Management Framework

Primary Objective: Traceability



Implication: Rigorous documentation is not optional; it is the license to operate.

Singapore emphasizes a centralized, positive approach to validation.

Governance Pipeline



Regulatory Framework: Centralized but Voluntary.
Model AI Governance Framework.

Key Guidelines: Positive, non-sanctioned approach.
Focus on implementation over penalties.

Primary Objective:
Technical Validation.



AI Verify Toolkit: **PASSED**

Governance Pipeline



PERSONA CARD

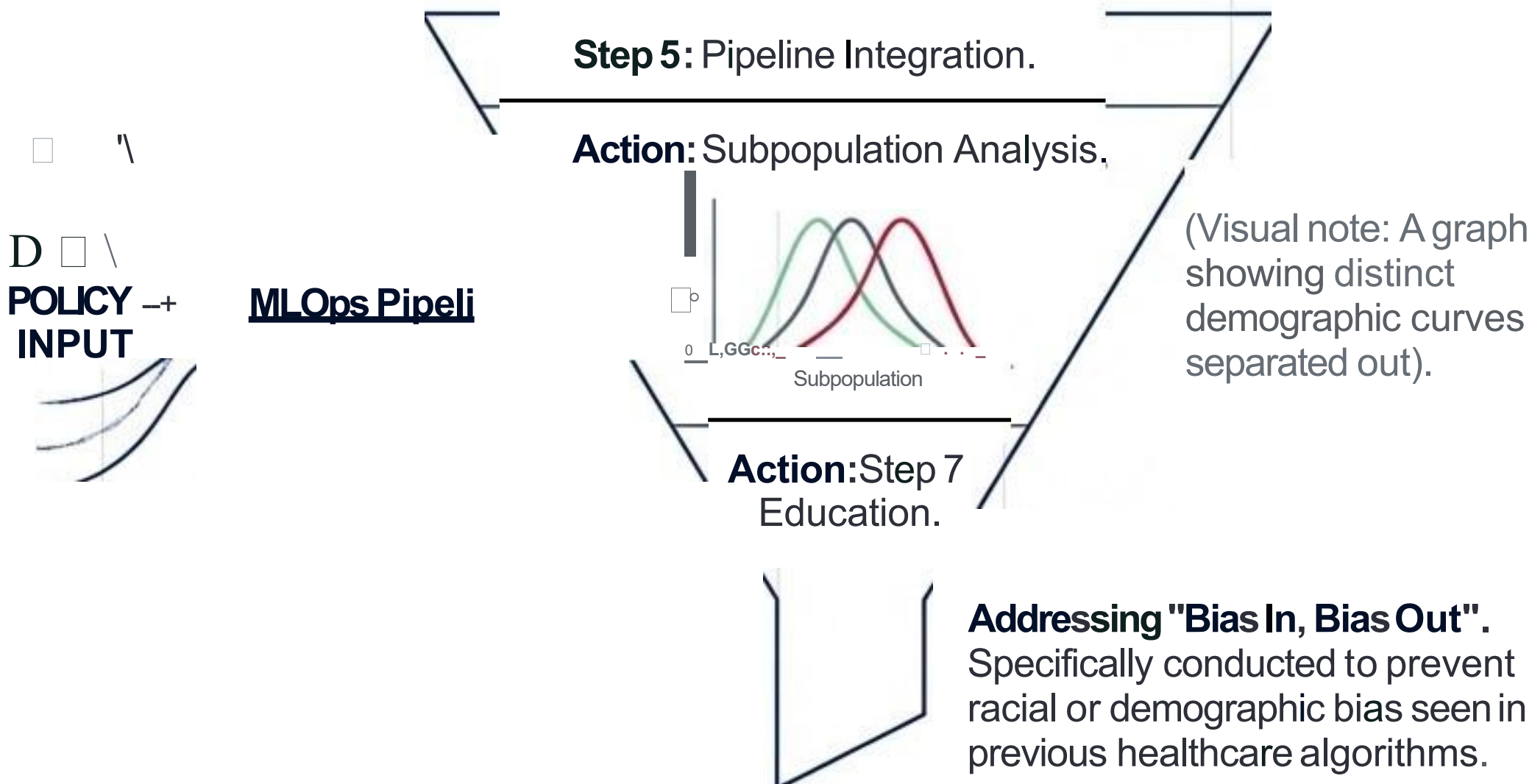
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Name: Data Scientist
Slate Grey (#4F5D75)

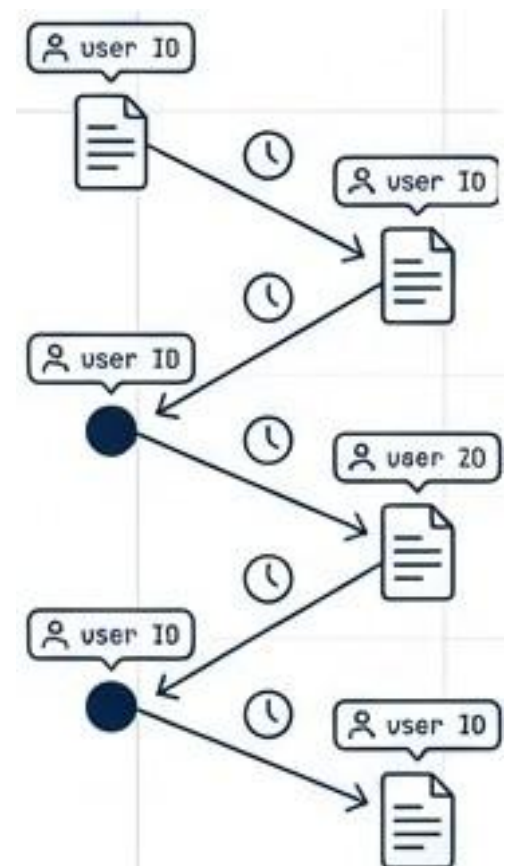
Merriweather:
Role: "The Data Scientist"
Focus: "Technical Execution"
Metric: "Robustness"

The Data Scientist translates policy into technical robustness to prevent bias.



We utilize "Glass Box" methods to ensure transparency and integrity.

Data Integrity (US ALCOA Principles)



Attributable: Traced to a specific individual or system.

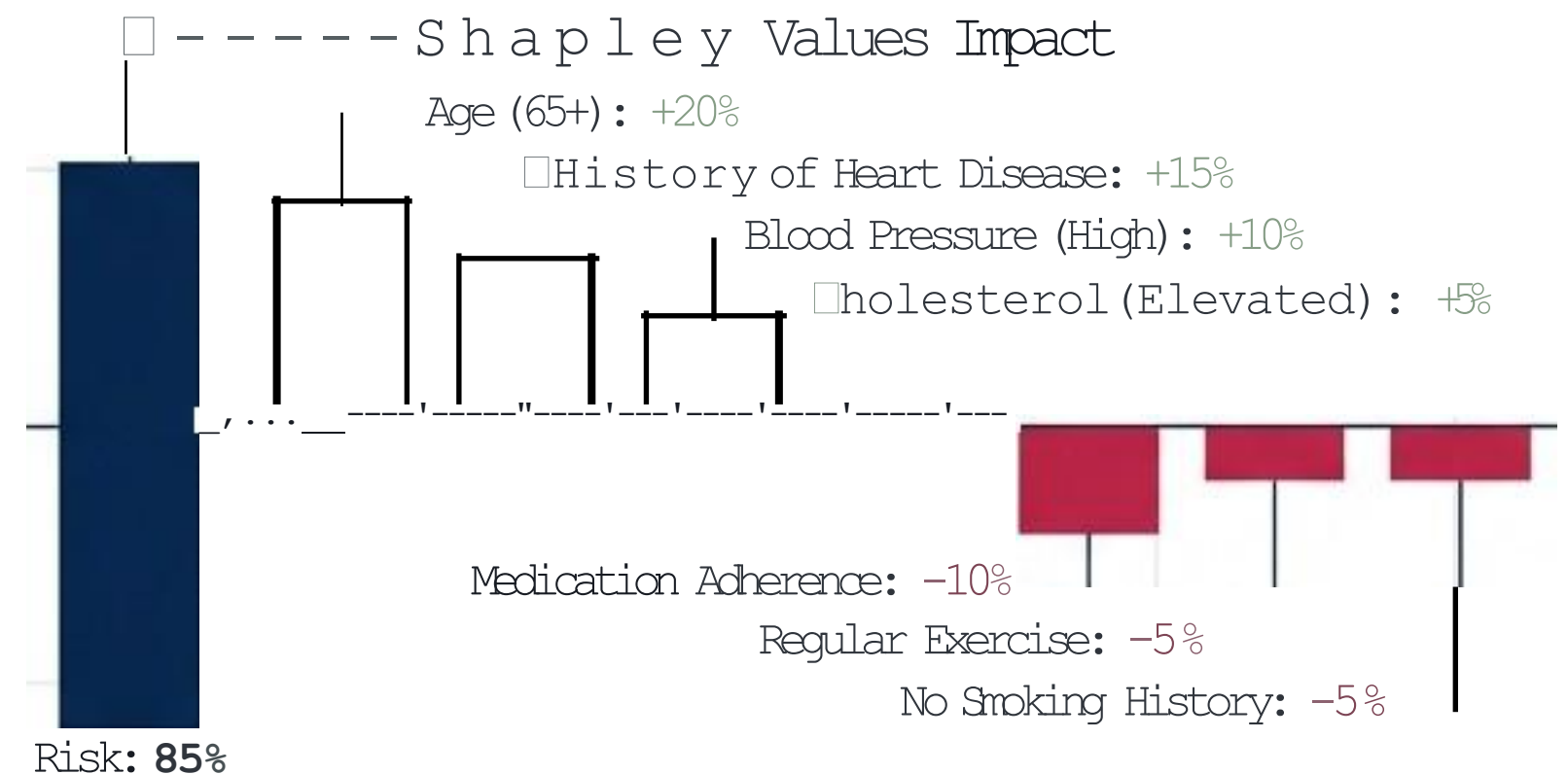
Legible: Permanent and readable.

Contemporaneous: Recorded at the time of execution.

Original: First record or certified copy.

Accurate: Error-free and reliable.

Prediction Understanding



Shapley Values allow a doctor to ask 'Why?' and validate the logic, showing how each feature contributes to the final prediction.

PERSONA CARD
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Name: Sage Precision
Slate Grey (#7DCEAO)

Role: IT / DevOps Lead
Focus: Sustainability & Monitoring
Metric: Resilience

The IT Lead ensures operational resilience and sustainability at scale.

Dashboard

0 CI/CD Gates

```
series {
  if (zevers:)
    reverts = code;
  else
    ceaktes = sss";
  data packets;
  ...
}
```

Merriweather:
Prevention of Knight Capital-style disasters.

f) Real-time Monitoring

A

Model Decay Check
JetBrains Hono

Merriweather:
Making governance the glue of the initiative without slowing down delivery.

B

Input Drift Check
JetBrains Mono



Automated metrics and gamification drive continuous compliance.

Gamification Strategy: Publishing KPI results incentivizes compliance through visibility. This reduces administrative burden and transforms governance from a blocker to an operational standard.

Step 8: Sustainability

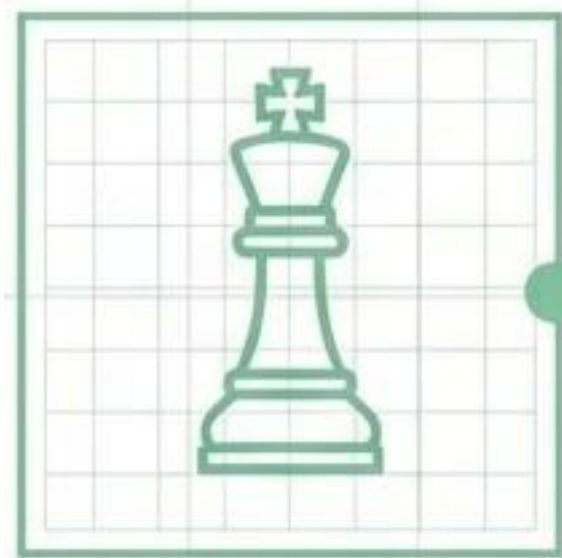
Leaderboard			
Team Member	Documentation	Drift Checks	Status
Dr. A. Sharma	98%	10/10	Compliant
J.Chen	100%	10/10	Compliant
M. Davis	95%	9/10	Compliant
L. Patel	99%	10/10	Compliant

A comparative summary of our dual-market regulatory strategy

United States 	Singapore 
Style: Helvetica Now Display Medium	Style: Helvetica Now Display Medium
Decentralized / Sectoral Merriweather Regular	Centralized / Voluntary Merriweather Regular
Focus:	Focus:
Accountability (GxP/ALCOA) Merriweather Regular	Practicality (Job redesign) Merriweather Regular
Tool: NIST AI Risk Management Framework JetBrains Mono 	Tool: AI Verify Toolkit JetBrains Mono 

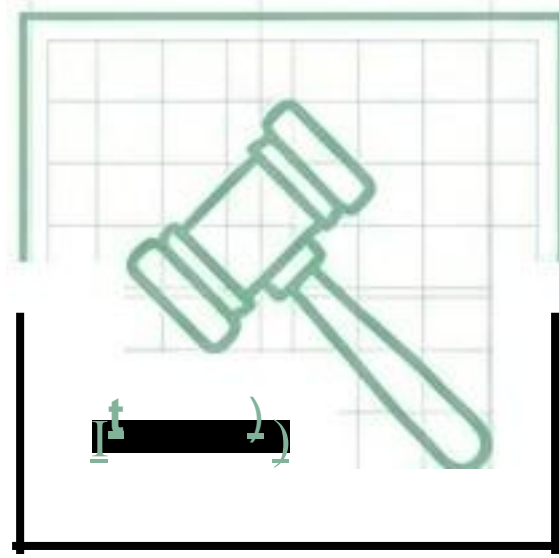
We successfully navigate both frameworks through a unified model registry.
Merriweather Regular

Integrated governance connects ethical strategy to daily operations.



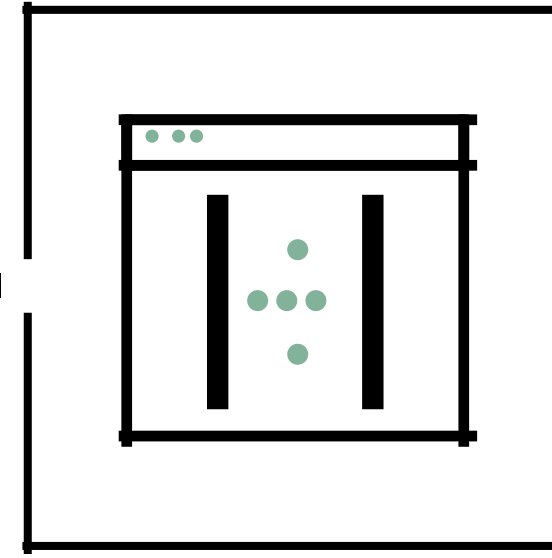
Strategy:

CEO defines Human-Centered Approach.
Merriweather Regular



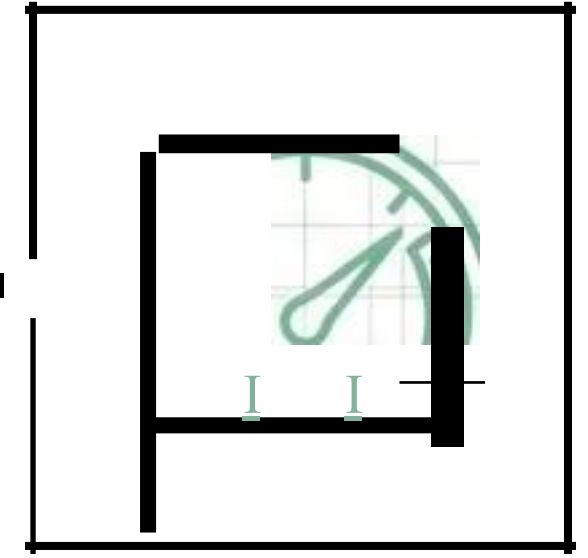
Policy:

COO selects High Criticality & Regional Rules.
Merriweather Regular



Execution:

Data Scientist codes Shapley Values.
Merriweather Regular



Monitoring:

IT Lead automates CI/CD Gates.
Merriweather Regular

This chain of custody ensures that the 'Opaque Technology Risk' is neutralized at every stage of the lifecycle.
Merriweather Regular

Governance is the glue that turns innovation into reliable healthcare.

Trust is our most valuable asset; without it, the technology is useless.

We have bridged the gap between US mandates and Singaporean pragmatism.

By gamifying and automating oversight, we ensure safety without sacrificing speed.

**Vital-Scan AI is ready for deployment:
Safe, Legal, and Ethical.**